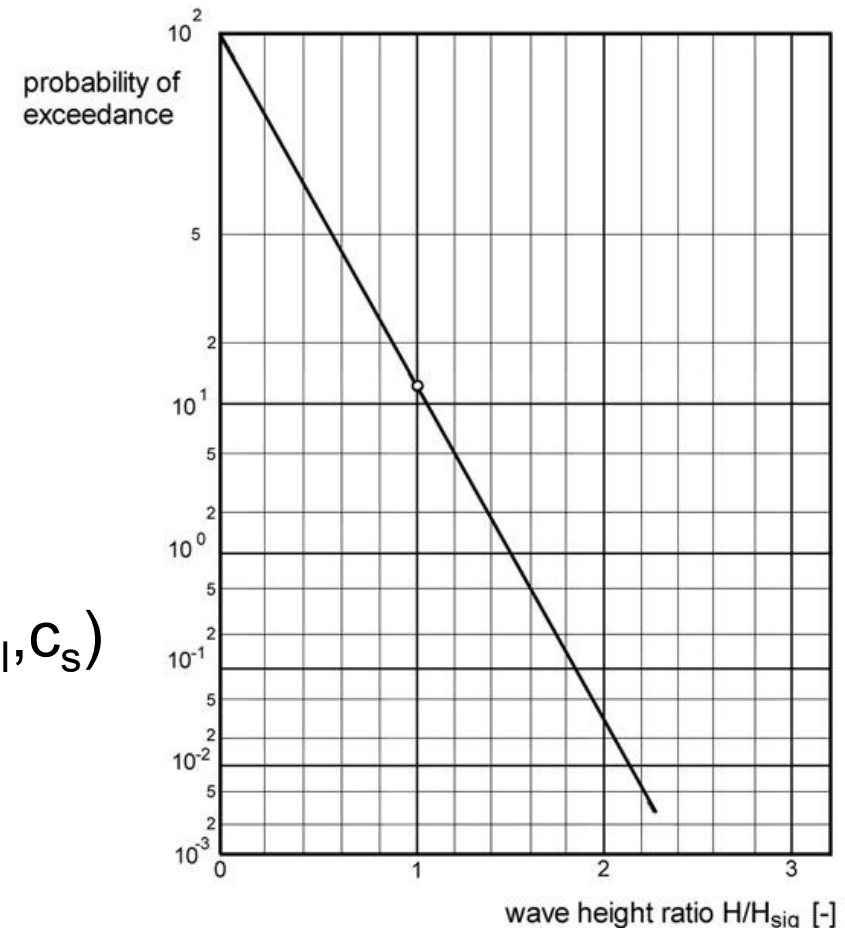


Shallow water

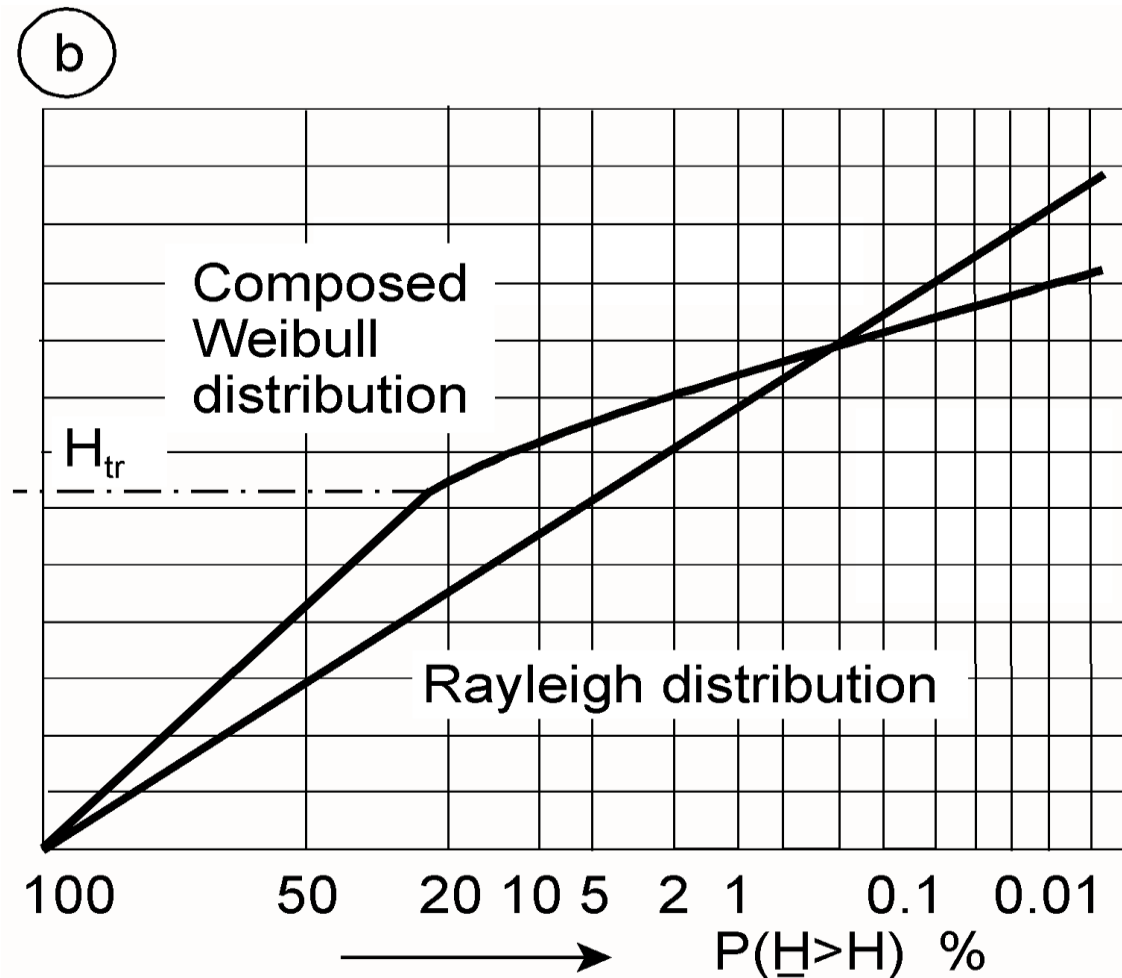
Problem

- VDM formula uses significant wave height H_s .
- Arguably, most damage is done by the highest waves, for instance $H_{2\%}$
- Van der Meer did his tests mainly with deep water waves, so the ratio $H_{2\%} / H_s = 1.4$ is fixed (Rayleigh dist.)
- The factor 1.4 is therefore implicitly present in the fit coefficients 6.2 and 1.0 (c_{pl}, c_s)
- Shallow water:
 - $h = 3H_{s,toe}$
 - $H_{s,toe} < 70\% H_{s,o}$



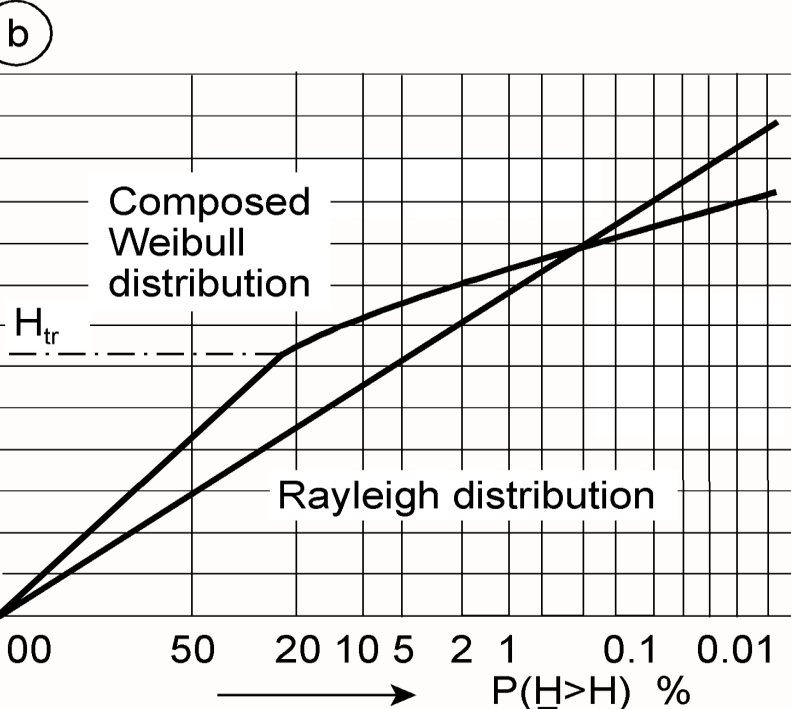
Problem

- In shallow water the Rayleigh distribution does not apply
- So this implicit assumption is not valid!



Short term wave statistics

Shallow water: **Composite Weibull distribution**
(Battjes and Groenendijk 2000)



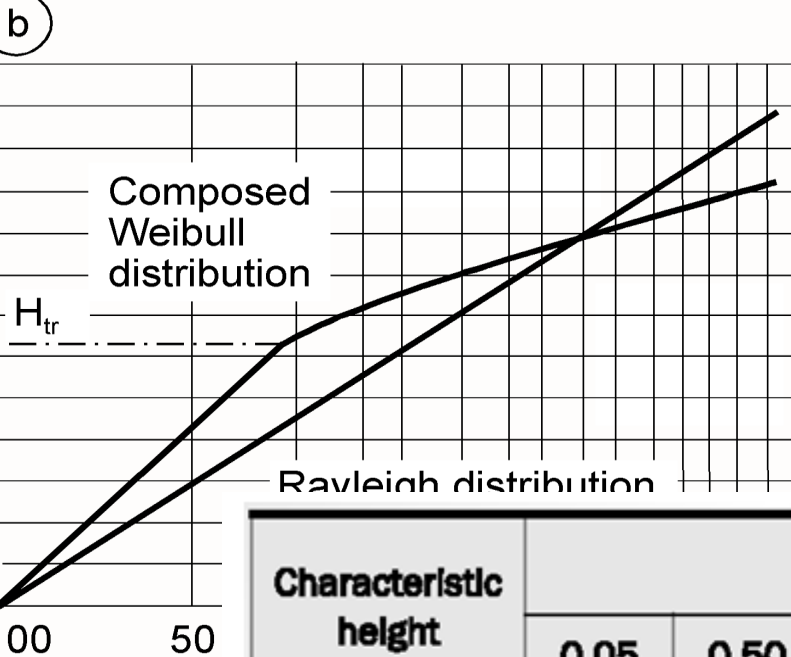
$$\Pr\{\underline{H} \leq H\} = \begin{cases} F_1(H) = 1 - \exp\left[-\left(\frac{H}{H_1}\right)^2\right] & H \leq H_{tr} \\ F_2(H) = 1 - \exp\left[-\left(\frac{H}{H_2}\right)^{3.6}\right] & H > H_{tr} \end{cases}$$

Procedure:

- Formula 1: Calculate H_{tr} from water depth and foreshore slope
- Formula 2: Calculate H_{rms} from H_{m0}
- Read results from table

Short term wave statistics

(Battjes and Groenendijk 2000)



- Rock Manual Box 4.4 (p359)
- Larger table in article

$$H_{tr} = (0.35 + 5.8 \tan \alpha) h$$

$$H_{rms} = \left[0.6725 + 0.2025 (H_{m0} / h) \right] H_{m0}$$

Characteristic height	Non-dimensional transitional wave H_{tr}/H_{rms}									
	0.05	0.50	1.00	1.20	1.35	1.50	1.75	2.00	2.50	3.00
$H_{1/10}/H_{rms}$	1.466	1.467	1.518	1.573	1.626	1.683	1.759	1.786	1.799	1.800
$H_{2\%}/H_{rms}$	1.548	1.549	1.603	1.662	1.717	1.778	1.884	1.985	1.978	1.978
$H_{1/3}/H_{rms}$	1.279	1.280	1.324	1.371	1.395	1.406	1.413	1.415	1.416	1.416

“Rewritten” Van der Meer formula

- Rewrite the formula accounting for shallowness of water
- Keep the ratio $H_{2\%}/H_s$ explicit

$$\frac{H_s}{\Delta d_{n50}} = c_{pl} \left(\frac{H_s}{H_{2\%}} \right) P^{0.18} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \xi_{0m}^{-0.5}$$

$$\frac{H_s}{\Delta d_{n50}} = c_s \left(\frac{H_s}{H_{2\%}} \right) P^{-0.13} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \sqrt{\cot \alpha} \xi_{0m}^P$$

- $c_{pl} = 6.2 * 1.4 = 8.7$
- $c_s = 1.0 * 1.4 = 1.4$
- Definition of ξ_{0m} has not changed
- In shallow water, calculate the actual $H_{2\%}/H_s$ ratio (e.g. using Battjes-Groenendijk method)
- Using H_s and the deep water formula is conservative

“Modified” Van der Meer formula

- The Rock Manual gives **another** modification of the Van der Meer formula for shallow water:
 - Re-calibrate coefficients on original dataset + new data with shallow water (Van Gent 2004)
 - Use $T_{m-1,0}$ instead of T_m

$$\frac{H_s}{\Delta d_{n50}} = c_{pl} \left(\frac{H_s}{H_{2\%}} \right) P^{0.18} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \xi_{m-1,0}^{-0.5}$$

$$\frac{H_s}{\Delta d_{n50}} = c_s \left(\frac{H_s}{H_{2\%}} \right) P^{-0.13} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \sqrt{\cot \alpha} \xi_{m-1,0}^P$$

- $c_{pl} = 8.4$; $\sigma_{cpl} = 0.7$;
 - $c_s = 1.3$; $\sigma_{cs} = 0.15$;
- Valid for $0.25 \leq H_{so}/h \leq 1.5$

Van Gent formula

- The Rock Manual also gives another formula (Van Gent et al. (2003)) based on:
 - No or not sufficient info about the period ($T_{m-1,0}$)
 - Independent of the wave period ($T_{m-1,0}$)
 - Independent of the ratio $\frac{H_{2\%}}{H_s}$

$$\frac{H_s}{\Delta d_{n50}} = 1.75 \sqrt{\cot \alpha} \left(1 + \frac{d_{n50,core}}{d_{n50}} \right) \left(\frac{S}{\sqrt{N}} \right)^{0.2}$$

- $\sigma = 0.10$ for rock slopes with a permeable core
 - $\sigma = 0.12$ for rock slopes with an impermeable core
- Permeability is accounted for via the ratio $\left(\frac{d_{n50,core}}{d_{n50}} \right)$
- Same for plunging and surging conditions
- Valid for $1.25 < h/H_{s-toe} < 3$

“Modified” Van Gent formula

- Scaravaglione et al. (2025) proposed the last formula for shallow water stability, considering very and extremely shallow water conditions:

$$\frac{H_{m0}}{\Delta d_{n50}} = 3.3\sqrt{\cot \alpha} \left(1 + \frac{d_{n50,core}}{d_{n50}} \right) s_{m-1,0}^{0.1} \left(\frac{S}{\sqrt{N}} \right)^{0.2}$$

- $\sigma = 0.36$ rock slopes with permeable core
- Small influence $s_{m-1,0}$
- Independent of the ratio $\frac{H_{2\%}}{H_s}$
- Uses spectral wave height (H_{m0}) instead of H_s
- Valid $0.4 \leq H_{m0,deep}/h \leq 5$